x	y	∇y	$\nabla^2 y$	$\nabla^3 y$
0	1			
		1		
1	2		1	
		2		0
2	4		1	
		3		0
3	7		1	
		4		0
4	11		1	
		5		0
5	16		1	
		6		0
6	22		1	
		7		
7	29			

To find y in terms of x $\therefore x_n + nh = x$, $x_n = 7, h = 1 \Rightarrow n = x - 7$.

$$y_{(x)} = 29 + (x-7)(7) + \frac{(x-7)(x-6)}{2}(1)$$

$$= 29 + 7x - 49 + \frac{1}{2}\left(x^2 - 13x + 42\right)$$

$$= \frac{1}{2}\left[58 + 14x - 98 + x^2 - 13x + 42\right]$$

$$= \frac{1}{2}\left[x^2 + x + 2\right]$$

5.2.4 Lagrange's interpolation formula

The Newton's forward and backward interpolation formulae can be used only when the values of x are at equidistant. If the values of x are at equidistant or not at equidistant, we use Lagrange's interpolation formula.

Let $y = f(x)$ be a function such that $f(x)$ takes the values $y_0, y_1, y_2, \ldots\ldots, y_n$ corresponding to $x = x_0, x_1, x_2 \ldots, x_n$ That is $y_i = f(x_i), i = 0,1,2,\ldots,n$. Now, there are $(n+1)$ paired values $(x_i, y_i), i = 0,1,2,\ldots,n$ and hence $f(x)$ can be represented by a polynomial function of degree n in x.

Then the Lagrange's formula is

$$y = f(x) = \frac{(x-x_1)(x-x_2)\ldots(x-x_n)}{(x_0-x_1)(x_0-x_2)\ldots(x_0-x_n)} y_0$$

$$+ \frac{(x-x_0)(x-x_2)\ldots(x-x_n)}{(x_1-x_0)(x_1-x_2)\ldots(x_1-x_n)} y_1 + \ldots +$$

$$\frac{(x-x_0)(x-x_1)\ldots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)\ldots(x_n-x_{n-1})} y_n$$

Example 5.22

Using Lagrange's interpolation formula find y(10) from the following table:

x	5	6	9	11
y	12	13	14	16

Solution:

Here the intervals are unequal. By Lagrange's interpolation formula we have

$$x_0 = 5, x_1 = 6, x_2 = 9, x_3 = 11$$

$$y_0 = 12, y_1 = 13, y_2 = 14, y_3 = 16$$

$$y = f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} \times y_0$$

$$+ \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} \times y_1$$

$$+ \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} \times y_2$$

$$+ \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} \times y_3$$

$$= \frac{(x-6)(x-9)(x-11)}{(5-6)(5-6)(5-11)}(12) +$$

$$\frac{(x-5)(x-9)(x-11)}{(6-5)(6-9)(6-9)}(13)$$

$$+ \frac{(x-5)(x-6)(x-11)}{(9-5)(9-6)(9-11)}(14) +$$

$$\frac{(x-5)(x-6)(x-9)}{(11-5)(11-6)(11-9)}(16)$$

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